

Writing Two-Step Equations from Real-World Contexts

Worksheet 7A.01.06 **ANSWER KEY**

For questions 1 and 2, name the known quantities and the unknown quantity. Then write an equation. Do not solve.

1. A town's water tower held 110,000 gallons and is draining at a rate of 2,150 gallons per hour. It now holds 79,900 gallons. How long has it been draining?

Known quantities. *110,000 gallons at the start; 2,150 gallons drained each hour; 79,900 gallons now.*

Unknown quantity. *h = the number of hours the tower has been draining.*

Equation. *$110,000 - 2,150h = 79,900$*

2. A giraffe born at a local zoo was 6 feet tall at birth and grows 0.08 foot each day during its first week. It is now 6.48 feet tall. How many days old is it?

Known quantities. *6 feet at birth; 0.08 foot gained each day; 6.48 feet now.*

Unknown quantity. *d = the number of days old the giraffe is.*

Equation. *$6 + 0.08d = 6.48$*

For questions 3 through 6, write an equation, solve it, and write one sentence saying what your solution means.

3. The average toddler grows 0.2 inch per month. James is 35.45 inches tall. If he grows at the average rate, how long will it take him to reach a height of 37 inches?

Equation. *$35.45 + 0.2m = 37$*

$0.2m = 1.55$

$m = 7.75$

Check: *$35.45 + 0.2(7.75) = 35.45 + 1.55 = 37$ ✓*

Meaning. *It will take James 7.75 months — about seven and three-quarter months — to reach 37 inches.*

0.2 is the rate, 35.45 is the starting value. Watch for students who divide 37 by 0.2 and never subtract.

4. Mr. Humphries has some quarters and four dimes in his pocket, worth \$2.40 in all. How many quarters does he have?

Equation. *$0.25q + 0.40 = 2.40$*

$0.25q = 2.00$

$q = 8$

Check: *$0.25(8) + 0.40 = 2.00 + 0.40 = 2.40$ ✓*

Meaning. *Mr. Humphries has 8 quarters.*

$25q + 40 = 240$ is equally correct — everything in cents instead of dollars. What is not correct is $25q + 40 = 2.40$, which mixes the two units.

5. A phone plan costs \$24.99 a month plus \$0.12 for each gigabyte of data used. Last month's bill was \$32.19. How many gigabytes were used?

Equation. $24.99 + 0.12g = 32.19$

$$0.12g = 7.20$$

$$g = 60$$

Check: $24.99 + 0.12(60) = 24.99 + 7.20 = 32.19$ ✓

Meaning. *60 gigabytes of data were used last month.*

The \$24.99 is charged whether any data is used or not, so it is the constant.

6. Four concert tickets were purchased. A \$2.50 service fee was added to each ticket. The total came to \$118.00. What was the price of one ticket before the fee?

Equation. $4(t + 2.50) = 118$

$$t + 2.50 = 29.50$$

$$t = 27.00$$

Check: $4(27 + 2.50) = 4(29.50) = 118$ ✓

Meaning. *One ticket cost \$27.00 before the service fee.*

The fee is charged on each of the four tickets, so the group repeats and needs parentheses. $4t + 2.50 = 118$ charges the fee once.

Questions 7 and 8 involve percentages. Read the order of the steps carefully.

7. Maddox has a coupon for \$10 off his purchase at Foot Zone. He found a pair of shoes marked 20% off. The store takes the 20% off first and then applies the coupon. Maddox paid \$36.20 before taxes. What was the original price of the shoes?

Equation. $0.80p - 10 = 36.20$

$$0.80p = 46.20$$

$$p = 57.75$$

Check: $0.80(57.75) = 46.20$, and $46.20 - 10 = 36.20$ ✓

Meaning. *The shoes were originally priced at \$57.75.*

Taking 20% off means paying 80%, so multiply by 0.80 — not by 0.20. The sale price is \$46.20, which is 80% of the original. The \$36.20 Maddox actually paid is about 63% of the original, because the coupon comes off after the percentage.

8. Victoria works as a salesperson. She earns \$350 each week plus a 10% commission on her weekly sales. This week her total earnings need to be \$800. How much does she need to sell?

Equation. $350 + 0.10s = 800$

$$0.10s = 450$$

$$s = 4,500$$

Check: $350 + 0.10(4,500) = 350 + 450 = 800$ ✓

Meaning. *Victoria needs to sell \$4,500 worth of product this week.*

Three different quantities live in this problem and they are easy to confuse: her sales are \$4,500, her commission is \$450, and her total earnings are \$800. The question asks for the sales.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. In each one the unknown appears on both sides of the equal sign. Write the equation first, then solve.

9. Gym A charges \$30 to join plus \$18 a month. Gym B charges \$75 to join plus \$12 a month. After how many months does a membership cost the same at both gyms?

Equation. $30 + 18m = 75 + 12m$

$$30 + 6m = 75$$

$$6m = 45$$

$$m = 7.5$$

Check: $30 + 18(7.5) = 165$, and $75 + 12(7.5) = 165$ ✓

Meaning. *After 7.5 months the two gyms cost the same — \$165 at each.*

Collect the m terms on one side first: $18m - 12m = 6m$. This is MA.8.AR.2.1 and it is the shape most Grade 8 equation items take.

10. Plan A costs \$45 a month plus \$0.10 for each gigabyte of data. Plan B costs \$30 a month plus \$0.25 for each gigabyte. At how many gigabytes do the two plans cost the same?

Equation. $45 + 0.10g = 30 + 0.25g$

$$45 = 30 + 0.15g$$

$$15 = 0.15g$$

$$g = 100$$

Check: $45 + 0.10(100) = 55$, and $30 + 0.25(100) = 55$ ✓

Meaning. *At 100 gigabytes the two plans both cost \$55.*

Moving the smaller g term keeps the coefficient positive. Below 100 GB Plan B is cheaper; above it, Plan A is.